FE Graded Quiz M3

Time left: 59:40

QUESTIONS

1	X	X	X
X	X	X	X
X	X	X	X
X	X	X	X

Question 1

Which method can we use to compare a distribution with a skew-normal distribution?

- Scatter plot
- QQ plot
- Residual plot
- Stem and Leaf plot

Question 2

A random variable X has the following possible outcomes and associated probabilities:

X	3	4	5	6	7
f(X)	0.1	0.2	0.2	0.3	0.2

Which of the following is the correct pair for expected value and variance?

- (5.00, 2.00)
- (5.70, 1.31)
- (5.30, 1.61)
- (4.50, 1,10)

Question 3

Which of the following steps is **not** part of the process for estimating a copula for continuous random variables X and Y?

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- Sample data points from a fitted copula function and compare with observed and transformed data.
- Use marginal CDFs to transform X and Y to uniformly distributed variables.
- Conduct goodness-of-fit tests after estimating the copula function.
- Select normal copula to estimate a copula model since normal copula accommodates any marginal CDFs.

Question 4

Which of the following statements is correct regarding the bias correction factor from concordance correlation?

- When the disagreement increases between two random variables, the bias correction factor gets larger.
- When the disagreement increases between two random variables, the standard deviations of two variables decrease.

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- When the disagreement increases between two random variables, the bias correction factor gets smaller.
- When the disagreement increases between two random variables, the standard deviations of the two variables also increases.

Question 5

For a symmetric distribution, the probability of X in the range of (mean - 2 X standard deviation, mean + 2 X standard deviation) is 90%. This distribution is

- Platykurtic
- Normal
- Leptokurtic
- Mesokurtic

Question 6

If we pick five cards with replacement from a deck of 52 playing cards, which of the following formulas expresses the probability of drawing two hearts?
(The deck is shuffled after replacement.)

- $\binom{5}{2}(0.5)^2(0.5)^3$
- $\binom{5}{2}(0.25)^2(0.75)^3$
- $\binom{5}{3}(0.5)^2(0.5)^3$

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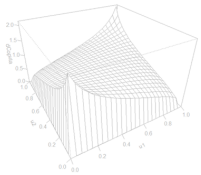
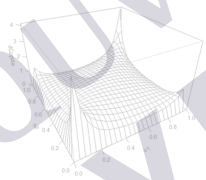
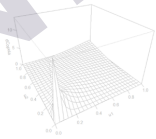
$\binom{5}{3}(0.25)^3(0.75)^2$

Question 7

Which of the following charts is related to Clayton copula?







Question 8

Two judges are deciding who is the best chef among five contestants. Two judges have ranked the chefs: The best is ranked 1, and the worst is ranked 5. Which of the following tables gives the correct concordant and discordant numbers?

	Judge 1	Judge 2	Concordant	Discordant
Chef 1	1	3	1	0
Chef 2	2	2	1	1
Chef 3	3	4	2	1
Chef 4	4	1	2	2
Chef 5	5	5		

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	Judge 1	Judge 2	Concordant	Discordant
Chef 1	1	3	4	1
Chef 2	2	2	3	1
Chef 3	3	4	3	0
Chef 4	4	1	2	0
Chef 5	5	5	1	0

	Judge 1	Judge 2	Concordant	Discordant
Chef 1	1	3	1	0
Chef 2	2	2	2	0
Chef 3	3	4	3	0
Chef 4	4	1	3	1
Chef 5	5	5	4	1

	Judge 1	Judge 2	Concordant	Discordant
Chef 1	1	3	2	2
Chef 2	2	2	2	1
Chef 3	3	4	1	1
Chef 4	4	1	1	0
Chef 5	5	5		

Question 9

What are the components in the centered parameter method for skew-normal distribution?

- Mean, Standard Deviation, Fisher-Pearson Standardized Moment Coefficient for Skewness
- Mean, Variance, Kurtosis
- Median, Variance, Fisher-Pearson Standardized Moment Coefficient for Skewness
- Median, Standard Deviation, Kurtosis

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Question 10

X is a standard normal random variable. What is the probability of $1 < X < 2$?

- 0.135
- 0.5
- 0.315
- 0.27

Question 11

Continuous random variables X and Y have a copula density function c_{xy}, marginal CDFs F(X),G(Y), and marginal PDFs f(X) and g(Y). X and Y's joint density function f(X,Y) is:

- $f(X,Y) = c_{xy}(F(X),G(Y))F(X)G(Y)$
- $f(X,Y) = c_{xy}(F(X),G(Y))f(X)g(Y)$
- $f(X,Y) = c_{xy}(f(X),f(Y))f(X)g(Y)$
- $f(X,Y) = c_{xy}(f(X),f(Y))F(X)F(Y)$

Question 12

Which of the following statistics is the most sensitive to outliers?

- Median
- Interquartile Range = 3rd quartile - 1st quartile

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- Mean
- Mode

Question 13

For a left skew distribution, what is the relationship among mean, median, and mode?

- Median < Mean < Mode
- Mean < Median < Mode
- Median > Mean > Mode
- Mean > Median > Mode

Question 14

Which of the following statements about a cumulative probability distribution function (CDF) is correct?

- The area under the CDF in a graph is 1
- The shape of a CDF in a graph can have a peak.
- The trend of a CDF in a graph is down from left to right
- If X is a random variable, its CDF is Prob (X <= x)

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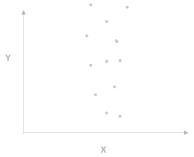
Question 15

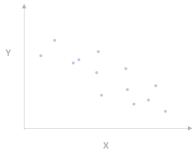
Which of the following equations does **not** meet the condition of invariance under monotonic transformation?

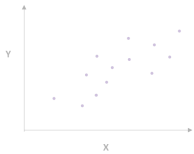
- C(X,Y) = C(2*X*, -2*Y*)
- C(X,Y) = C(2*X*,2*Y*)
- C(X,Y) = C(exp(X), ln(Y))
- C(X,Y) = C(ln(X),ln(Y))

Question 16

We have two random variables X and Y, and their Pearson correlation = -0.7. Which of the following plots is most likely to be their scatter plot?







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